

Multiple Choice

1. **Answer:** A

Options: A) Golgi apparatus and endoplasmic reticulum; B) Lysosomes and endosomes; C) Plasma membrane and cytoskeleton; D) DNA and mRNA; E) Proteins and lipids

Note: Correct option: A - Golgi apparatus and endoplasmic reticulum

2. **Answer:** C

Options: A) They can reproduce asexually.; B) They can produce viable gametes through mitosis.; C) They have a relative evolutionary fitness of zero.; D) Their offspring are always genetically identical to the parents.; E) They cannot produce offspring due to lack of genetic variation.

Note: Correct option: C - They have a relative evolutionary fitness of zero.

3. **Answer:** A

Options: A) Acetylcholine will not be degraded in the synaptic cleft.; B) Acetylcholine will not bind to receptor proteins on the postsynaptic membrane.; C) Acetylcholine will not be produced in the presynaptic membrane.; D) Acetylcholine will be excessively produced in the synaptic cleft.; E) Acetylcholine will not be transported to the synaptic cleft.

Note: Correct option: A - Acetylcholine will not be degraded in the synaptic cleft.

4. **Answer:** A

Options: A) Both species benefit equally; B) One species benefits, and the other is significantly harmed; C) One species harms the other without any benefit; D) Both species harm each other; E) The species exchange benefits, but one benefits more than the other

Note: Correct option: A - Both species benefit equally

5. **Answer:** A

Options: A) 4; B) 16; C) 32; D) 64; E) 192

Note: Correct option: A - 4

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The incidence of CPM is not increased in IVF/ICSI pregnancies compared with spontaneous conceptions. CPM probably does not account for the adverse perinatal outcomes following IVF/ICSI.

7. **Answer:** False

Options: A) True; B) False

Note: The effect on venous pressures caused by the change in patient positioning alone during liver surgery does not affect the risk of venous air embolism.

8. **Answer:** True

Options: A) True; B) False

Note: The incidence of post-tonsillectomy late haemorrhage in our study population was 1.78%. A statistically significant difference was found between night-time and day-time haemorrhages. Even though no significant distribution of haemorrhages per hour was observed, we underline that we recorded 32 (54.2%) events in 2 periods of the day: from 10 p.m. to 1 a.m. and from 6 to 9 a.m.

9. **Answer:** True

Options: A) True; B) False

Note: It is important to continue FA supplementation over the long term in patients on methotrexate and FA in order to prevent them discontinuing treatment because of mouth ulcers or nausea and vomiting. Our data suggest that FA supplementation is also helpful in preventing neutropenia, with very little loss of efficacy of methotrexate.

10. **Answer:** True

Options: A) True; B) False

Note: The presence of a high signal intensity change on T 2-weighted MRI and the pyramidal tract sign can be used as prognostic factors for patients with CSA.

Long Form Response

11. **Answer:** Sickle-cell disease is more deadly in America. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Only animals and plants have basic nutritional requirements.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key